



## RELAZIONE PERIODICA

Mesi di maggio-giugno 2026

|                              |                                                                                                  |
|------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Dottorando</b>            | Ken Tanaka Hernandez                                                                             |
| <b>CUP progetto</b>          | I53C21000370006                                                                                  |
| <b>Corso di dottorato</b>    | Applied Data Science and Artificial<br>Intelligence (ADSAI)<br>Università degli Studi di Trieste |
| <b>Ciclo-anno accademico</b> | XXXIX                                                                                            |
| <b>Tutor scientifico</b>     | Monica Sugan                                                                                     |

### ATTIVITÀ DI RICERCA E FORMATIVE TUTORAGGIO E DIDATTICA CONFERENZE RISULTATI DELLA RICERCA ALTRO

Refactored for catalog production and HPC execution. Changes introduced lazy date-indexed catalogs, event/pick loading, spatial filtering, plotting, BGMA comparison, QC, utilities and tests; expanded data, downloader, station and magnitude support; and replaced the Leonardo workflow with parameterized indexer, picker, associator and NLL locator jobs. Configuration added velocity-model and stage mappings. Final detail work added INSTANCE PhaseNet support. The merge integrated both branches and was counted once.

#### Conferences:

Carducci, A., Cocina, O., Sciotto, M., Cannata, A., Di Gioia, S., Vuan, A., Saraò, A., Tanaka Hernández, K., and Sugan, M.: Performance assessment of Deep Learning picking models at

Mount Etna volcano, EGU General Assembly 2026, Vienna, Austria, 3–8 May 2026, EGU26-11585, <https://doi.org/10.5194/egusphere-egu26-11585>, 2026.

#### Papers:

Carducci, A., O. Cocina, M. Sciotto, A. Cannata, S. Di Gioia, A. Vuan, A. Saraò, K. Tanaka Hernández, and M. Sugan (2026). Insights into a Deep Learning Workflow for Automatic Detection of Seismic Signals at Mount Etna Volcano, *Seismol. Res. Lett.* doi: 10.1785/0220250431.

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Lomax A., Virieux J., Volant P., Berge-Thierry C. (2000) Probabilistic Earthquake Location in 3D and Layered Models. In: Thurber C.H., Rabinowitz N. (eds) *Advances in Seismic Event Location. Modern Approaches in Geophysics*, vol 18. Springer, Dordrecht. [https://doi.org/10.1007/978-94-015-9536-0\\_5](https://doi.org/10.1007/978-94-015-9536-0_5)

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Michelini, A., Cianetti, S., Gaviano, S., Giunchi, C., Jozinović, D., and Lauciani, V.: INSTANCE – the Italian seismic dataset for machine learning, *Earth Syst. Sci. Data*, 13, 5509–5544, <https://doi.org/10.5194/essd-13-5509-2021>, 2021.

Mousavi, S. M., Ellsworth, W. L., Zhu, W., Chuang, L. Y., & Beroza, G. C. (2020). Earthquake transformer—An attentive deep-learning model for simultaneous earthquake detection and phase picking. *Nature Communications*, 11(1), 1–12. <https://doi.org/10.1038/s41467-020-17591-w>

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Zhu, W., & Beroza, G. C. (2019). Phasenet: A deep-neural-network-based seismic arrival-time picking method. *Geophysical Journal International*, 216(1), 261–273. <https://doi.org/10.1093/gji/ggy423>

Zhu, W., McBrearty, I. W., Mousavi, S.M., Ellsworth, W. L., & Beroza, G. C. (2022). Earthquake phase association using a Bayesian Gaussian Mixture Model. *Journal of Geophysical Research: Solid Earth*, 127, e2021JB023249. <https://doi.org/10.1029/2021JB023249>



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Sperimentale

| Dottorando   | Tutor scientifico                                                                                   |
|--------------|-----------------------------------------------------------------------------------------------------|
| <i>Firma</i> | <i>Firma</i><br> |

Data: 20 giugno 2026